

DOCUMENTATION

Use Case 1: Borrow Equipment

Primary Actor	Student
Preconditions	Student and equipment must exist in the system.
Main Action	Student submits a request to borrow a specific piece of equipment.
Expected Result	The borrowing request is validated, recorded, and marked as active, while the equipment is set to unavailable.
Possible Failure	Student is restricted, equipment does not exist or is unavailable, or student has reached max active borrowings.

Use Case 2: Return Equipment

Primary Actor	Student
Preconditions	An active borrowing record exists for the student and equipment.
Main Action	Student returns the borrowed item back to the laboratory.
Expected Result	The borrowing record status updates to returned, and the equipment becomes available again.
Possible Failure	Borrowing record is not found or is already marked as returned.

Use Case 3: Find Available Equipment

Primary Actor	Student
Preconditions	System contains equipment records.
Main Action	Student requests a list of equipment currently available for borrowing.

Expected Result	System displays a filtered list of all items marked as available.
Possible Failure	No equipment matches the query or system storage is unreachable.

Domain Concept	Information / Properties Contained	Rules & Managed State	Not Responsibility Of
Student	• Student ID • Name • IsAllowedToBorrow (Bool) • Active Borrowings Count	• Must be active to borrow • Cannot exceed maximum active borrowing limit	• Direct database access • Managing equipment inventory • Rendering UI forms
Equipment	• Equipment ID • Name / Model • IsAvailable (Bool)	• Can only be borrowed if IsAvailable is true • Toggles availability on borrow/return	• Validating student eligibility • Storing borrowing history
Borrowing	• Borrowing ID • Student ID • Equipment ID • Date Borrowed • Expected Return Date • Borrowing Status (Enum)	• Tracks borrow lifecycle (Active / Returned) • Calculates return dates	• Direct database persistence • Printing user notifications